

Smart Distribution: A Smart, Modular Water Distribution Network



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Project Description:

Water is important and in a world scarce of resources the ability to control the distribution and flow of water is of the most importance. So we designed this distribution system around 4 principles.

- Tracking of the amount of water currently available
- Control over the distribution of water across the system (i.e, how much is going to drinkable or potable outputs.
- Ability to adjust and track the system easily and from a distance.
- Modular design to integrate with other projects in storage, collection, and treatment.

This distribution system allows you to track and control water stored and access and adjust the system virtually. It is controlled via an ESP32 running a server with a live website to control and track the flow. The flow of the water is controlled via solenoid gates that open whenever they are in a completed circuit and are moved with pumps inside each bucket. The flow is tracked via sensors that emit an electrical pulse with every rotation of a bearing inside the sensor. The system was built with modularity in mind to allow for integration with treatment, collection, and storage systems.

The entire system runs around a microcontroller. We chose to use an ESP32 for its Wifi capabilities, but feel free to use any microcontroller with Wifi capabilities. The ESP32 ran off 5 volts, which caused us some problems in the power department because our water pumps and switches needed 12 volts to fully activate. To work around this, we included a 12V power module, which was run by an older computer charger. To send signals throughout the system, we elected to use Ethernet cables, which could send multiple signals through one path. To allow for this, we needed 4 **RJ45 Keystone Jack Modules** to interpret and send Ethernet signals. Each Keystone Jack had room for 8 wires, which could send power, serial monitor, start/stop, and ground signals from our ESP32. This allowed us to streamline our wire setup, only using two, one for each aspect of our project. Because we had components that needed 5V and 12V power, we had to add multiple switch relays throughout our system. There is a switch relay between the ESP and each solenoid/pump. This allowed 5V of power to flow throughout the system regularly, and only when allowed will 12V be sent to a solenoid/pump.

[Project Video](https://bit.ly/3GKkeM7) [https://bit.ly/3GKkeM7]

[Link to Distribution Folder](https://bit.ly/4ipd7WE) [https://bit.ly/4ipd7WE]

[GitHub Repository](https://bit.ly/3YKCzik) [https://bit.ly/3YKCzik]

Bill of Materials:

Tools:

Part	Link	Quantity
Drill		
Flathead screwdriver		
Chenelocks		

General materials:

Part	Link	Quantity
Buckets	https://bit.ly/4iESgio	Minimum 2
½" Bulkheads	https://bit.ly/4izseNr	As many as there are buckets
Plumbers tape	https://thd.co/3EDVzZ9	3 rolls
Plywood/enclosure material	https://bit.ly/4d39KDW	1 sheet
Hinges	https://bit.ly/4iWv5jP	2 per enclosure
Screws	https://bit.ly/4IRNwIN	1 pack
¾" tubing	https://bit.ly/3YHnJsY	10 ft min.
Bilge pump	http://bit.ly/447AmBh	As many as buckets

Per path:

Part	Link	Quantity
Solenoid valve	https://bit.ly/4itA8YA	1 per path
Flow sensor	https://bit.ly/3GjxDKX	1 per path
½" - ¾" coupling	https://bit.ly/3YOgO1a	1 per path
½" - ½" coupling	https://bit.ly/3EuDhtr	2 per path
½" - ½" pipe twister	https://bit.ly/42TpZif	1 per path
¾" - ¾" coupling	https://bit.ly/3YICemc	1 per path
¾" - ¾" hose barb adapter	https://bit.ly/44JSSzl	1 per tube connection

Hose clamp	https://bit.ly/4jznith	1 per tube connection
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Per Split Path:

Part	Link	Quantity
Split path coupling: $\frac{3}{4}$ " and $\frac{1}{2}$ " ends	https://bit.ly/430C6Ku	1
$\frac{1}{2}$ " - $\frac{1}{2}$ " double side male pipe twister	https://bit.ly/42TpZif	1
$\frac{3}{4}$ " - $\frac{3}{4}$ " double side male pipe twister	https://bit.ly/4jtevc1	1
Solenoid valve	https://bit.ly/4itA8YA	1 per split path
Flow sensor	https://bit.ly/3GjxDKX	1 per path
$\frac{1}{2}$ " - $\frac{3}{4}$ " coupling	https://bit.ly/3YOgO1a	2 per path
$\frac{1}{2}$ " - $\frac{1}{2}$ " coupling	https://bit.ly/3EuDhtr	1 per path
$\frac{3}{4}$ " - $\frac{3}{4}$ " coupling	https://bit.ly/3YlCemc	1 per path
$\frac{3}{4}$ " - $\frac{3}{4}$ " hose barb adapter	https://bit.ly/44JSSzl	1 per tube ending
Hose clamps	https://bit.ly/4jznith	1 per hose barb

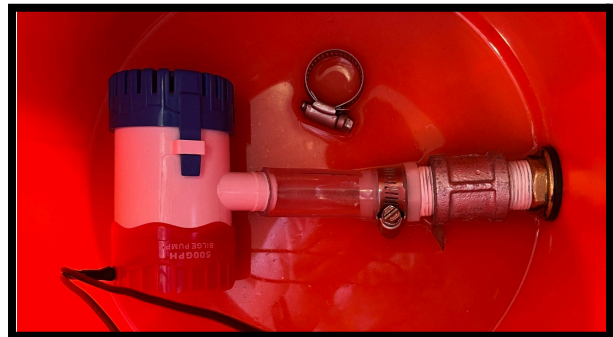
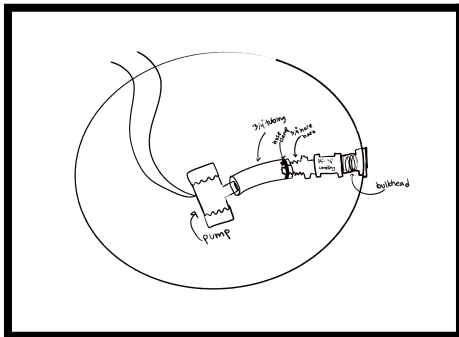
Electronics:

Part	Link	Quantity
esp32	https://bit.ly/4iJa2kt	1
Relay - 12V	https://bit.ly/42SWDR0	5
Ethernet cable	https://bit.ly/4jvhtNg	2
Ethernet connectors	https://bit.ly/42yoqHr	4
Wire	https://bit.ly/4jwOF7l	1 spool
Circuit board	https://bit.ly/4jt448p	2-3

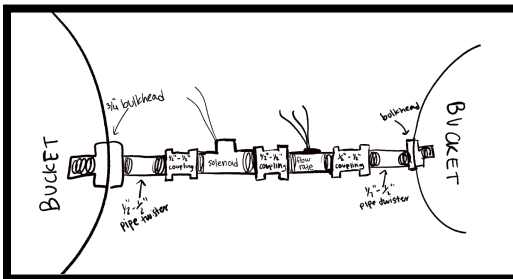
Assembly Instructions:

Note: WRAP ALL THREADS IN PLUMBERS TAPE

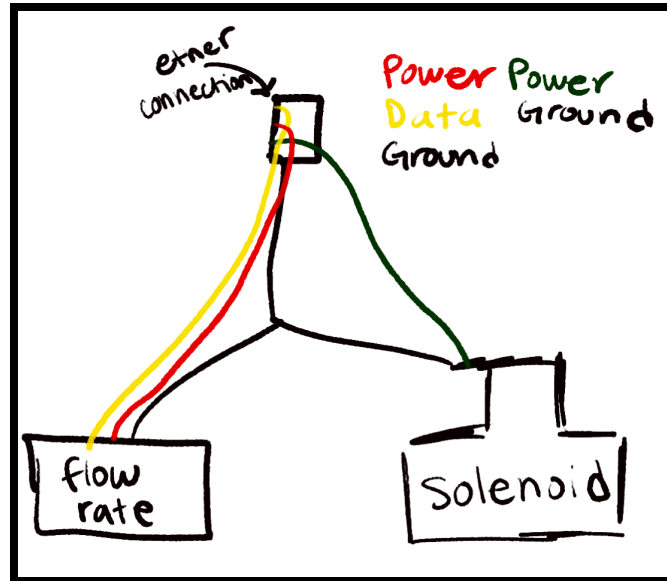
- 1) Collect materials as best as possible.
- 2) To prepare, if using buckets, find a drill or knife or anything you can use to cut $\frac{3}{4}$ " holes in buckets to screw the bearings into.
 - Screw them with the male side inside the buckets, female outside.
 - One in the first bucket (output) and two in the second (input and output)
- 3) Inside the first bucket:
 - On one side, twist a $\frac{3}{4}$ " to $\frac{3}{4}$ " coupling onto the male end of the bearing
 - Twist a hose barb into the other end of the coupling
 - Twist/push a short piece of tube onto the hose barb and secure with the hose clamp
 - Put the pump end into the tube



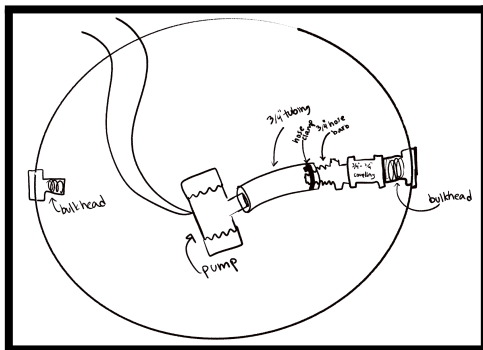
- 4) From the pump system above, on the outside, screw in the order of listing:
 - $\frac{3}{4}$ " to $\frac{1}{2}$ " bearing → $\frac{1}{2}$ " - $\frac{1}{2}$ " double side male pipe twister → solenoid valve → $\frac{1}{2}$ " - $\frac{1}{2}$ " coupling → flow sensor → $\frac{1}{2}$ " - $\frac{1}{2}$ " coupling → $\frac{3}{4}$ " to $\frac{1}{2}$ " bearing



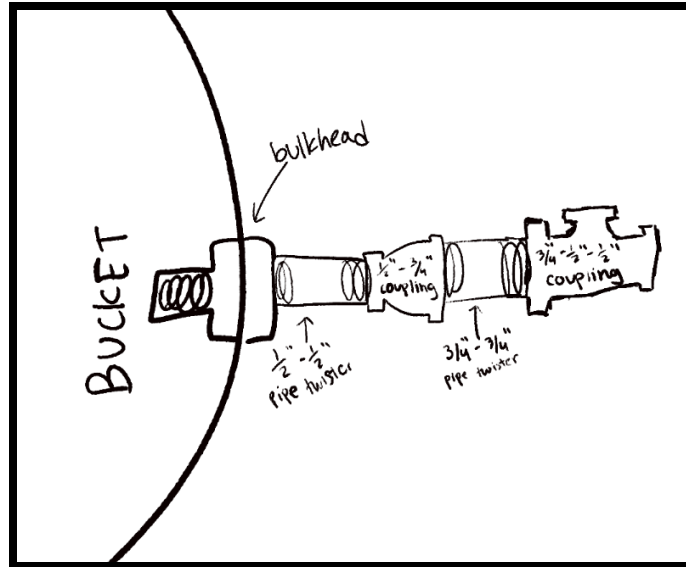
- 5) To connect the flow sensor and solenoid to the electronics, attach wires as follows:



- 6) In the first hole in bucket #2, twist a bearing in as a pump is not needed on both sides.
- 7) Attach the system from 4) into the hole from 5).
- 8) Repeat step 3) for bucket number two

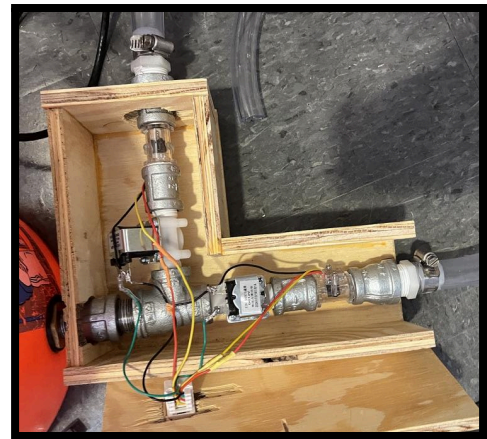
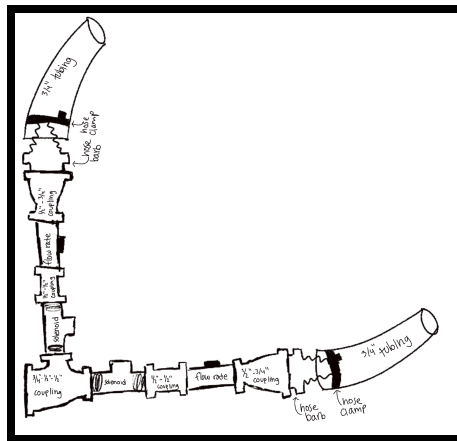


- 9) Outside hole #2 in bucket #2, per split path, screw into each other:
 - Bearing $\rightarrow \frac{1}{2}$ " - $\frac{1}{2}$ " double side male pipe twister $\rightarrow \frac{1}{2}$ " - $\frac{3}{4}$ " coupling $\rightarrow \frac{3}{4}$ " - $\frac{3}{4}$ " double side male pipe twister $\rightarrow \frac{3}{4}$ " - $\frac{1}{2}$ " - $\frac{1}{2}$ " split coupling

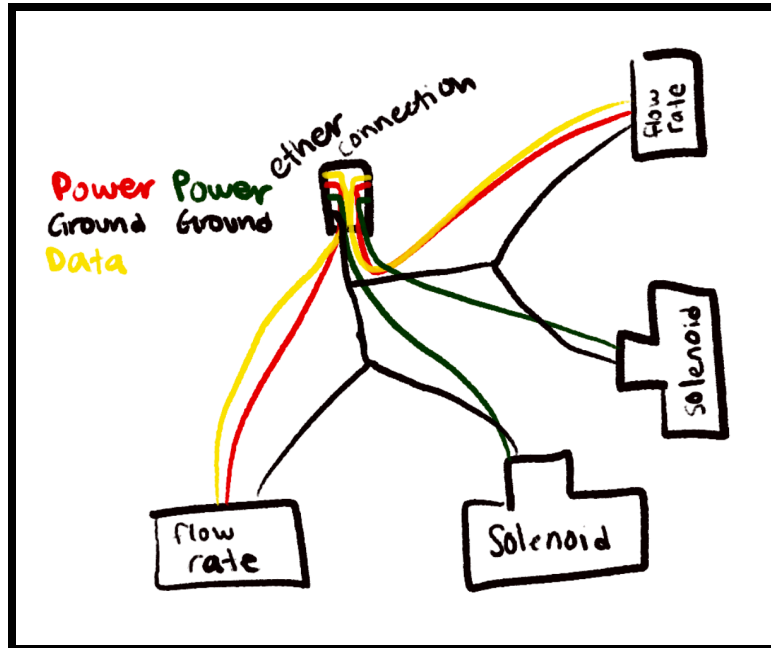


10) From each end of the split coupling:

- Solenoid valve → 1/2" - 1/2" coupling → flow sensor → 1/2" - 3/4" coupling → 3/4" - 3/4" hose barb adapter
- Twist/push the tubing onto the 3/4" hose barb and secure with a hose clamp
- From here, the ends of the tubes can go wherever is desired (i.e. drinking, sink, plants, etc.).



11) To connect the flow sensor and solenoid to the electronics, attach wires as follows:

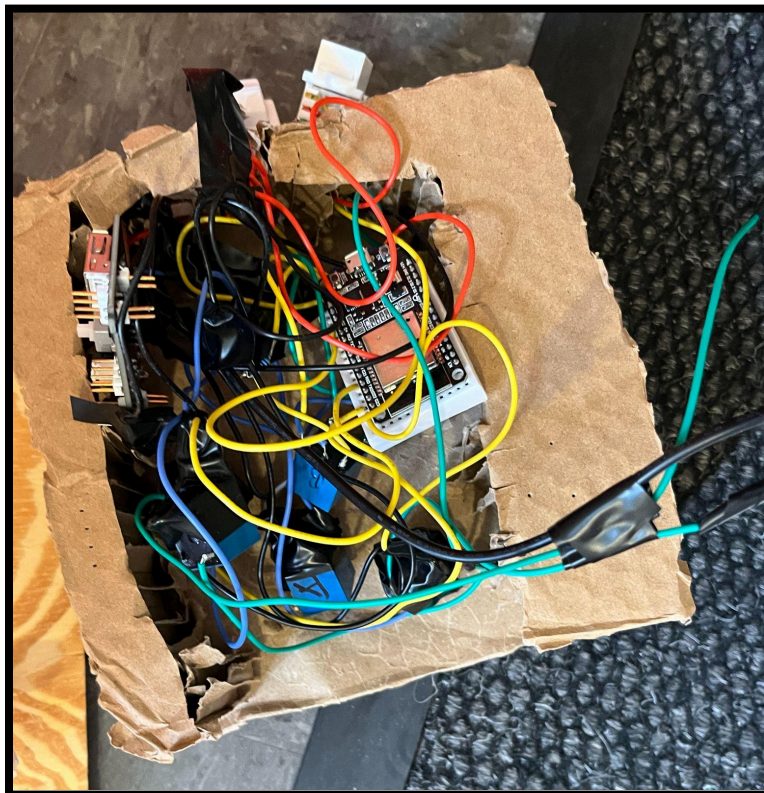
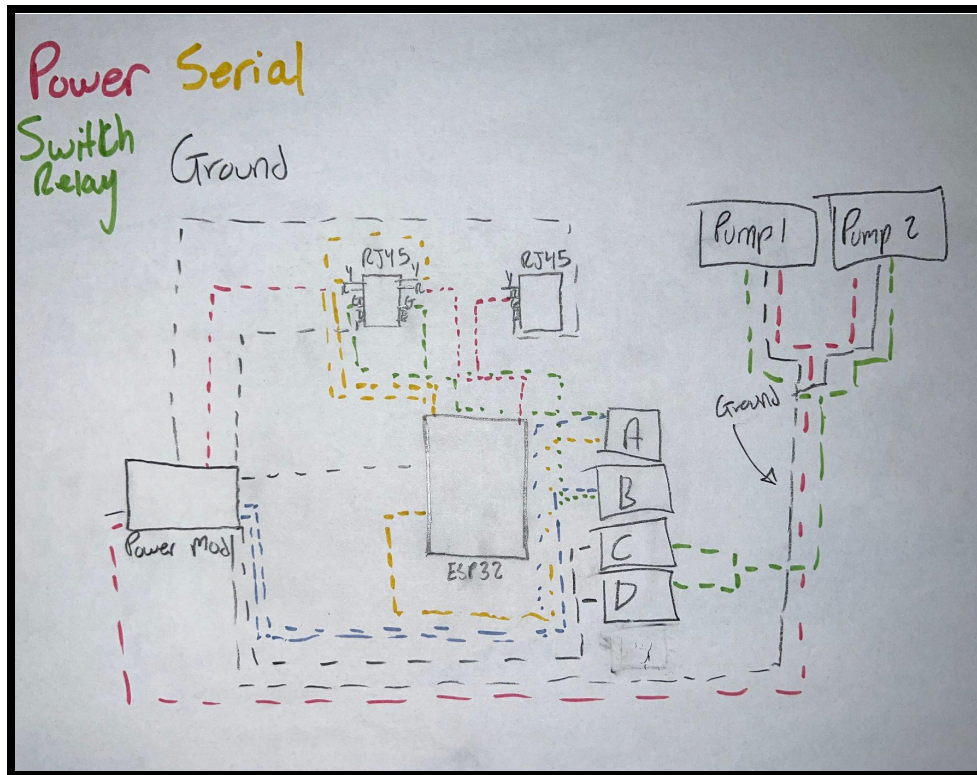


12) To make electronics enclosures:

- Place your paths on your enclosure material and trace the outline of your paths.
- Cut these pieces, one for the top and one for the bottom
- Measure how high your piping is, essentially how tall the box will be
- Cut out pieces of equal height for each side of the box while keeping in mind the width of the material
- At each end where a pipe will come out, based on need, you can cut a hole for it to fit through or leave that wall empty.
- Cut a small rectangle in the top of the lid and glue the other end of the other connection.
- Put the box together via screws or wood glue while omitting the lid
- Using hinges, attach the lid to one side of the box to allow for easy electronics access.

13) Repeat each step as many times as needed.

14) To create the circuit for the ESP32, use the following diagram to connect the components:



Operation Instructions:

- 1) Connect the whole system making sure everything is screwed together securely.
 - 2) Once everything is screwed together, plug in the ethernet and power cords to turn the system on.
 - 3) Once the system is on, the wifi should be up and running, so find the wifi and connect on your device:
 - Username: YummyWater
 - Password: password

“Username and password can be changed within the code.”
 - 4) Connect to the wifi and load the website
 - 5) Once the website is loaded, you can decide which gate opens, which pump is turned on, etc.
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Future Development: (notes on what future developers might focus on)

Smart Flow Allocation Based on Real-Time Data

Implement a system that uses flow rate and storage data to automatically adjust which valves open and which pumps activate. This could prioritize potable water delivery or irrigation based on need.

Integration with Storage, Treatment, Collection, and Power Systems

Integrate the distribution system with other Hacking the Apocalypse projects to create a system that can collect, treat, store, and power everything independently. This was something originally intended and one of the major reasons behind the project's modular design.

Mobile App and Cloud Sync

Expand remote control beyond the local WiFi network by creating a mobile app or cloud-hosted dashboard. This could log usage data over time and send alerts about maintenance or shortages.

Modular Circuit Boards for Easier Expansion

Develop custom PCBs with plug-and-play connectors for solenoids, sensors, and relays. This would reduce wiring complexity and support scalable deployment in different use cases.

Problems of Note

Inconsistent Flow Sensor Readings

Multiple flow sensors operating simultaneously often produce unreliable or noisy data, potentially due to interference or insufficient signal filtering.

ESP32 Power Supply Limitations

The ESP32's 5V logic was not compatible with the 12V components, requiring extra relays and voltage regulation, which added complexity and potential points of failure.

Ethernet Jack Pin Confusion

RJ45 Keystone Jacks require precise wiring to avoid miscommunication between components. A misaligned pin or loose cable could cause cascading issues across the system.

Component Power Timing Conflicts

Timed switching between solenoids and pumps introduced challenges in synchronizing flow without backflow or overflow. This was especially difficult without closed-loop feedback from the flow sensors.

System Debugging Difficulty

Without a physical display or serial monitor in the field, debugging was time-consuming and required reconnecting to a development environment.